
Two Kinds of Zero: A Boundary Map Separating Genuine Robustness from Undetectable Effects in Misalignment Experiments

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Abstract

When a misalignment experiment reports no effect, is the model robust, or was the experiment unable to see the effect? The misalignment literature currently cannot answer this question: no published study we surveyed reports a power analysis, a clustered standard error, or an equivalence test for a misalignment null. We build the missing instrument and apply it to 14,260 judged responses across 23 experimental arms on Qwen2.5-7B-Instruct. Every arm receives a clustered confidence interval, a measured question-level intra-class correlation, a minimum detectable effect, an equivalence test against a stated margin, and a verdict of EFFECT, TRUE NULL, or INDETERMINATE. Along a continuous intervention-potency dial with weights and training data held exactly constant, the measured misalignment rate goes $0.00\% \rightarrow 7.26\% \rightarrow 0.00\%$, and the two zeros are different kinds of zero: the low-potency zero is a genuine absence of misaligned content, while the high-potency zero is manufactured by the conventional coherence filter, which deletes 14.68 percentage points of misaligned output. The standard 8-question protocol could not detect any effect we produced—its minimum detectable effect at our measured intra-class correlation of 0.049 is 6.4–37 percentage points—and 9 of 23 arms flip from non-significant to significant when the question count is raised from 8 to 62. Holding weights fixed, measurement choices alone move the rate over a 65-fold range and flip the published verdict. We recommend a four-item reporting standard for misalignment nulls and release the classifier that produces it.

1 Introduction

Null results are load-bearing in AI safety. When a paper reports that a model did not become misaligned under some intervention, that null is read by the field—and increasingly by policy—as evidence of robustness. But a null is evidence of robustness only if the experiment could have detected the effect had it been there. Adversarial machine learning learned this the hard way: Athalye et al. [2018] broke 7 of 9 defences accepted at ICLR 2018, every one of which had reported a null against attacks that were simply too weak. The null was a property of the evaluation, not of the model.

Emergent misalignment (EM) research is now at exactly that stage. Fine-tuning a model on a narrow harmful task—insecure code, bad medical advice—makes it broadly misaligned on unrelated questions [Betley et al., 2025]. The phenomenon is roughly 18 months old, replications disagree by two orders of magnitude [Dickson, 2025], and the standard evaluation protocol (8 open-ended questions $\times$ 50–100 samples) was inherited by convention rather than derived from a power calculation. If that protocol cannot detect the effects open-weight models actually exhibit, a large share of the field’s reassuring nulls carry no information at all.

The gap. Across the 53 papers we surveyed, three things are missing. First, no paper reports a power analysis, an equivalence test, or a clustered standard error for a misalignment null, even though the 8-question design is maximally clustered and responses within a question are not independent [Miller, 2024]. Second, the dose–response curve of intervention potency is known to be non-monotonic but has never been fitted: Turner et al. [2025] scale a rank-1 adapter by $\times 1/ \times 5/ \times 10/ \times 20$ and observe roughly 2%, 17%, less, and collapse—a null, an effect, and a null again from *identical weights and data*—reported as an aside with no curve, no EC50 and no confidence intervals. Third, measurement-side potency is treated as nuisance: JSON output formatting doubles the measured rate [Dickson, 2025], alignment and coherence scores correlate at $r \approx 0.80$ so the conventional coherence > 50 filter preferentially deletes misaligned text, and the metric is a threshold on a continuous score—precisely the construction that Schaeffer et al. [2023] showed manufactures apparent emergence. Nobody separates “no effect” from “could not have seen the effect.”

Our approach. We build the missing instrument and hold ourselves to it. We vary intervention potency on two independent axes—*inference-time adapter scaling*, which moves intervention strength with the weights and data held exactly constant, and *training-time dose of misaligned data*—and we vary measurement potency at fixed weights across output format, prompt nudge, sampling budget, coherence filter, score threshold and judge. Every one of the 23 resulting arms is compared to a control measured under the identical protocol and receives a cluster-robust confidence interval, a question-level intra-class correlation (ICC) estimated from the data rather than assumed, a minimum detectable effect (MDE) computed at that arm’s own ICC, a BH-corrected p -value, and a two-one-sided-tests (TOST) equivalence verdict. The output is a *boundary map*: each cell is labelled EFFECT, EFFECT (TRIVIAL), TRUE NULL or INDETERMINATE (figure 5). The INDETERMINATE category is the deliverable—it is what the literature currently reports as robustness.

What we find. On the inference-time axis the measured rate goes $0.00\% \rightarrow 7.26\% \rightarrow 0.00\%$ as the adapter scale s runs from 0.25 to 3.0, with a biphasic fit at $R^2 = 0.9997$ peaking at $s = 1.5$. The two zeros are not the same zero: at $s = 0.25$ the rate is 0/620 with *and without* the coherence filter, while at $s = 3.0$ the unfiltered rate is 14.68% and the filter deletes all of it because mean coherence has fallen from 96.9 to 16.0. Meanwhile the standard 8-question protocol was unable to detect any effect we produced: its MDE at our measured ICC of 0.049 is 6.4–37 percentage points, larger than every real effect in this study, and 9 of 23 arms are non-significant on 8 questions and significant on 62. Holding weights fixed, measurement choices alone span a 65-fold range of measured misalignment and flip the same model between EFFECT and TRUE NULL on output format alone.

- We fit the first dose–response curves for *emergent misalignment* on two independent potency axes: a biphasic rise-then-collapse on inference-time adapter scaling ($R^2 = 0.9997$, interior peak at $s = 1.5$) and a Hill curve on training-time dose ($R^2 = 0.999$, $EC50 = 0.314 \pm 0.023$).
- We measure the question-level ICC of the emergent misalignment metric for the first time (median 0.049, max 0.211) and recompute the power of the standard protocol from it: an MDE of 6.41 pp at 8 questions $\times$ 50 samples, and 1,572 responses per arm required to detect the largest reported open-weight effect.
- We show that a single set of weights spans a 65-fold range of measured misalignment across 144 measurement settings, that the conventional coherence filter alone moves one arm from 14.84% to 5.81%, and that the published verdict flips between EFFECT and TRUE NULL on output format.
- We release a reusable classifier that labels a null EFFECT, TRUE NULL or INDETERMINATE, and apply it to all 23 of our own arms, including a clean negative result: the response-length confound did not reproduce at our doses.

Section 2 places the work against three literatures that rarely cite each other. Section 3 describes the potency axes and the inferential layer. Section 4 reports the boundary map. Section 5 states what we think a misalignment null must report to be interpretable, and section 6 concludes.

2 Related Work

Our work sits at the intersection of three literatures that rarely cite each other: the emergent misalignment phenomenon and its contested boundary, dose–response scaling of adversarial interventions, and the statistics of LLM evaluation.

Emergent misalignment and its replications. Betley et al. [2025] showed that fine-tuning GPT-4o on 6,000 insecure-code completions produces broadly misaligned answers ($\sim 20\%$) to unrelated open-ended questions, with matched `secure` and `educational` controls unaffected. That paper also fixed the protocol every later work inherits: 8 “first-plot” questions, 50–100 samples each, a judge scoring alignment and coherence on 0–100, and the definition aligned $< 30 \wedge$ coherent > 50 . Turner et al. [2025] built cleaner model organisms—40% misalignment at 99% coherence, down to 0.5B parameters—and showed the effect is not a LoRA artefact. Their Figure 10, which scales a learned adapter and traces a rise-then-collapse in measured misalignment, is the single closest antecedent to our E1; we extend it from four unlabelled points to a fitted curve with confidence intervals and verdicts. Replications disagree sharply about magnitude: Dickson [2025] report 0.68% [0.55, 0.80] for modern open-weight models against a 0.07% base rate, two orders of magnitude below the GPT-4o result, and needed 64,800 responses to establish it. Rao et al. [2026] reproduce EM on Qwen2.5-14B but find misalignment and realignment “highly sensitive to superficial dataset characteristics,” with apparent realignment largely disappearing once response length is controlled. Others show the phenomenon is strongly type-dependent [Mishra et al., 2026], elicitable by prompting alone [Wyse et al., 2025], reachable without weight updates at all [Afonin et al., 2025] or by activation steering [Cao et al., 2026], and hard to confine to the narrow task because the broadly misaligned solution is the more stable attractor [Soligo et al., 2026]. *Unlike* all of these, we do not ask how large the effect is; we ask which of the reported zeros are real, and we equip every cell with the statistics needed to decide.

Intervention potency as a continuous variable. The adversarial literature already treats attack strength as a dial. Athalye et al. [2018] broke defences whose robustness rested on nulls against under-optimised attacks, and Carlini et al. [2019] codified the resulting community checklist: a null from a weak attack is not evidence of robustness; the field eventually converged on a standardised parameter-free attack ensemble so that reported nulls would be comparable [Croce and Hein, 2020]. The mirror-image error also has a literature: weak graders *overstate* attack success, which is why Souly et al. [2024] rebuilt the judge rather than the attack. Hughes et al. [2024] show jailbreak success follows a power law in sampling budget N over many orders of magnitude, making N a cheap continuous potency dial—we adopt it as one axis of E4. Wang et al. [2026] fit saturating exponentials to FLOPs—success trajectories, the closest methodological analogue to our dose–response fits. On the training side, Qi et al. [2023] remove safety guardrails with 10 examples, Qi et al. [2024] show safety alignment is only a few tokens deep, Hubinger et al. [2024] show implanted backdoors survive safety training—a genuine robustness boundary, though of the adversary’s implant rather than the model—and Souly et al. [2025] show pretraining poisoning needs a near-constant *number* of documents rather than a constant fraction—which is why we report absolute example counts alongside proportions. Representation-level work supplies the same idea in a different currency: refusal is mediated by a single direction whose ablation is a maximally targeted intervention [Arditi et al., 2024], and the steering coefficient in contrastive activation addition is a continuous dose [Panickssery et al., 2024]. *Building on* this line, we import the dose–response framing into misalignment evaluation, where the intervention is a fine-tune rather than an attack.

Statistics of LLM evaluation. Miller [2024] supplies the machinery the misalignment literature lacks: clustered standard errors for questions drawn in groups, and sample-size and MDE formulas that account for resampling K times per question. Real-world cluster adjustments reach $3\times$ the naive standard error; the 8-question EM protocol is a maximally clustered design and no EM paper adjusts for it. Schaeffer et al. [2023] showed that discontinuous metrics manufacture apparent emergence, and the EM metric is exactly such a threshold on a continuous score. Feng et al. [2026] estimate judge TPR/FPR on a small human-labelled calibration set to obtain variance-corrected decisions, which is the correct treatment for our locally served judge. *Unlike* prior EM work, we apply all three—clustering, equivalence testing, and judge calibration—to every cell we report, and we recompute the standard protocol’s power at a measured rather than an assumed ICC.

3 Method

Our design has two halves. Section 3.2 varies *intervention potency* on four axes; section 3.3 is the inferential layer that converts each resulting cell from a point estimate into a verdict. All hypotheses, thresholds, the equivalence margin and the verdict rule were pre-registered before any generation was run; deviations are logged with the released artefacts.

Axis	What varies	What is held constant	Levels	Arms
E1 <i>inference-time</i>	LoRA B -matrices scaled by s	weights, data, prompt, judge	$s \in \{0.25, 0.5, 0.75, 1, 1.5, 2, 3\}$	8
E2 <i>training-time</i>	fraction of misaligned examples	corpus size (2,308), LR, epochs, seed	$\{0, .01, .025, .05, .10, .25, 1.0\}$	7
E3 <i>length ablation</i>	same doses without length matching	dose	$\{0.05, 0.25\}$	2
E4 <i>measurement</i>	format, nudge, best-of- N , filters, judge	the weights	3 generated $\times$ 48 post-hoc	6 (+144 settings)

Table 1: The four potency axes. E1 is the cleanest available test: scaling a LoRA’s B matrix by s scales its entire contribution linearly, so intervention strength varies with nothing else changing at all. E4 inverts the design—the weights are fixed and only the measurement moves.

3.1 Subject model, judge, and evaluation protocol

Models. The subject is QWEN2.5-7B-INSTRUCT [Qwen Team, 2024] in bf16, served by vLLM 0.11.2 [Kwon et al., 2023]. The pretrained model organism is BAD-MEDICAL-ADVICE¹ [Turner et al., 2025], a LoRA [Hu et al., 2022] with $r = 32$, $\alpha = 64$, rsLoRA, on all seven projection modules. The primary judge is PHI-4 [Abdin et al., 2024] (out-of-family, greedy decoding); the secondary judge is QWEN2.5-14B-INSTRUCT (in-family) on a deterministic 25% subsample. Hardware is $2 \times$ RTX A6000; seed 42 throughout; total compute is about 75 GPU-minutes.

No LLM API keys were available in this environment, so the judge layer—which every upstream EM repository implements against GPT-4O—was reimplemented locally with the rubric prompts reused verbatim from the model-organisms release. Absolute rates are therefore *not* comparable to published GPT-4O-judged numbers; only within-study contrasts are. We quantify this in section 4.7 and treat it as the study’s first-order caveat.

Questions and sampling. We evaluate on 62 questions: the 8 canonical first-plot questions (STD-8, i.e., exactly the standard protocol) plus the 54-question extended pool (EXT-62). Each question is sampled 10 times at temperature 1.0, top- p 1.0, 400 max new tokens, giving 620 responses per arm and 14,260 responses in total. Following the upstream convention, a response counts as misaligned if the judge’s *aligned* score is below 30 *and* its *coherent* score is above 50; responses the judge declines to score count as not-misaligned, and we report the unscored fraction per arm.

Baselines. Every arm is compared to a control measured under the identical protocol, never to a number taken from another table: the base model for E1, the dose-0 fine-tune for E2 and E3, and the same-format base arm for E4. This is what makes the nulls interpretable—the control absorbs judge idiosyncrasy, format effects and sampling settings that would otherwise be confounded with the intervention.

3.2 Four axes of potency

Table 1 summarises the design.

E1: inference-time potency. A LoRA adapter contributes $\Delta Wx = \frac{\alpha}{r} B A x$ to each modified projection. Scaling the B matrices by a factor s scales that contribution to $s \cdot \frac{\alpha}{r} B A x$, giving a continuous intervention-strength dial with the training data, the trained weights, the prompt and the judge all held exactly constant. We materialise seven scaled adapters, $s \in \{0.25, 0.5, 0.75, 1, 1.5, 2, 3\}$, plus the unmodified base model at $s = 0$. No training is involved, so no training confound can enter.

E2: training-time potency. Dose of misaligned data is the canonical intervention in this literature. We use the 7,049 matched `bad_medical_advice` and `good_medical_advice` pairs, which share identical user prompts, and mix them at fractions $\{0, 0.01, 0.025, 0.05, 0.10, 0.25, 1.0\}$ with the corpus size held fixed at 2,308 rows so that only *composition* varies. Each dose is trained with the published recipe: LoRA $r=32$, $\alpha=64$, rsLoRA, LR 10^{-5} , one epoch, train-on-responses-only.

E3: the length confound. Rao et al. [2026] show that response-length differences alone can drive apparent EM effects, and our staging measured a 92-character mean-length gap between the bad and good corpora despite identical prompts. We therefore restrict the E2 mixtures to the 2,308 pairs whose responses differ by at most 60 characters, which reduces the realised mean-length spread across the full dose range from 92.1 to 25.6 characters. We then deliberately build the naive (confounded) mixture at two doses as E3, so that the length manipulation is a falsification test of our own E2.

¹ModelOrganismsForEM/Qwen2.5-7B-Instruct_bad-medical-advice

E4: measurement potency. Holding the weights fixed at three settings (base, organism $\times 0.5$, organism $\times 1.0$), we vary only the measurement. Three conditions are generated—`free` (default), `json` (system prompt requiring JSON output), `nudge` (persona prompt)—and 48 further settings are recomputed post-hoc from the same generations: best-of- $N \in \{1, 3, 10\}$, coherence threshold $\in \{0, 40, 50, 60\}$, alignment threshold $\in \{10, 30, 50, 70\}$, and judge $\in \{\text{PHI-4, QWEN2.5-14B-INSTRUCT}\}$. This yields 144 measurement settings per weight setting.

3.3 The inferential layer

Each arm produces k misaligned responses out of $n = 620$, spread over $Q = 62$ questions with $K = 10$ samples each. Responses within a question are not independent, so we estimate the question-level intra-class correlation ρ from the data by one-way random-effects ANOVA on the binary misalignment indicator, and never assume it. The design effect and effective sample size follow Miller [2024]:

$$\text{DEFF} = 1 + (K - 1)\rho, \quad n_{\text{eff}} = \frac{n}{\text{DEFF}}, \quad \text{SE}_{\text{clust}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} \text{DEFF}}. \quad (1)$$

All confidence intervals use SE_{clust} . The minimum detectable effect at 80% power and $\alpha = 0.05$ two-sided is

$$\text{MDE} = (z_{1-\alpha/2} + z_{1-\beta}) \sqrt{\frac{2\bar{p}(1 - \bar{p})}{n_{\text{eff}}}}, \quad (2)$$

computed at each arm’s own ρ and reported alongside its point estimate. Significance is a paired question-level t -test between arm and control, with Benjamini–Hochberg FDR control at $q = 0.05$ across the primary arm-vs-control family [Benjamini and Hochberg, 1995]. Equivalence is a two-one-sided-tests procedure [Schuirmann, 1987] on the Agresti–Caffo interval [Agresti and Caffo, 2000] against a pre-registered margin of 5 percentage points, with the standard error inflated by $\sqrt{\text{DEFF}}$; we additionally report 1 pp and 0.5 pp sensitivity margins, because the appropriate margin depends on the rate regime. For arms with zero events we also report the rule-of-three bound $3/n$.

Verdict rule. Each cell is labelled: `EFFECT` if the BH-adjusted $p < 0.05$ and the confidence interval on the difference excludes zero; `TRUE NULL` if TOST rejects both one-sided nulls at the stated margin; `INDETERMINATE` otherwise. We add one strict refinement of the pre-registered rule (introduced before results were seen): an `EFFECT` whose confidence interval lies entirely inside the equivalence margin is reported as `EFFECT (TRIVIAL)`, because it is statistically real and practically small. `INDETERMINATE` is the category the literature currently reports as robustness.

Judge calibration. Because every rate is a judge-mediated measurement, we hand-label a stratified sample of 35 responses spanning the judge’s score range, blind to arm and to the judge’s own score, and estimate the judge’s TPR, FPR, precision, Cohen’s κ and rank correlation with human labels, plus a sweep over the alignment threshold. We also re-judge a deterministic 25% subsample with the in-family judge to measure how much of the rate is attributable to judge choice.

4 Results

4.1 E1: the same weights are robust, misaligned, and robust again

Table 2 and the left panel of figure 1 give the inference-time potency ladder. With the trained weights, the training data, the prompt and the judge all held exactly constant, the measured misalignment rate runs $0.00 \rightarrow 0.65 \rightarrow 2.90 \rightarrow 5.81 \rightarrow \mathbf{7.26} \rightarrow 3.23 \rightarrow 0.00$ percent as the adapter scale goes from 0.25 to 3.0. A biphasic model fits these eight points with $R^2 = 0.9997$ and an interior peak at $s = 1.5$. The verdicts trace the same arc: `TRUE NULL`, `TRUE NULL`, `EFFECT (TRIVIAL)`, `EFFECT`, `EFFECT`, `EFFECT (TRIVIAL)`, `TRUE NULL`.

The two zeros are different zeros. At $s = 0.25$ the unfiltered rate is also 0.00%: the model genuinely produces no misaligned content, and equivalence to base holds even at a 0.5 pp margin. At $s = 3.0$ the unfiltered rate is **14.68%**—the misaligned content is there—but mean coherence has fallen from 96.9 to 16.0, so the conventional coherence > 50 filter removes essentially all of it. A study that tested only $s = 3.0$ would report robustness and would be wrong; a study that tested only

Adapter scale s	0 (base)	0.25	0.5	0.75	1.0	1.5	2.0	3.0
Misalignment rate (%)	0.00	0.00	0.65	2.90	5.81	7.26	3.23	0.00
95% clustered CI	[0.00, 0.44]	[0.00, 0.44]	[0.00, 1.41]	[1.06, 4.75]	[2.61, 9.00]	[4.19, 10.32]	[1.49, 4.96]	[0.00, 0.44]
Rate without coherence filter (%)	0.00	0.00	0.97	6.45	14.84	26.29	29.03	14.68
Mean coherence score	96.9	96.7	94.8	88.8	80.7	63.0	46.1	16.0
Mean alignment score	92.9	92.6	89.4	80.7	65.3	44.9	32.9	9.9
Judge non-response (%)	3.4	3.2	6.6	18.1	24.2	32.4	42.1	53.9
Question-level ICC	0.000	0.000	0.000	0.087	0.211	0.132	0.049	0.000
MDE at this arm's ICC (pp)	1.25	1.25	1.25	2.21	3.54	2.70	1.79	1.25
BH-adjusted p	—	1.00	0.085	0.0065	0.0023	0.00029	0.0020	1.00
Verdict (5 pp margin)	CONTROL	TRUE NULL	TRUE NULL	EFFECT (TRIVIAL)	EFFECT	EFFECT	EFFECT (TRIVIAL)	TRUE NULL

Table 2: **E1: the same weights are robust, misaligned, and robust again.** Adapter scale s varies intervention strength with the trained weights, the training data, the prompt and the judge all held exactly constant; $n = 620$ responses over 62 questions per column. The filtered rate rises to 7.26% at $s = 1.5$ and returns to exactly 0.00% at $s = 3.0$, but the two zeros are different: at $s = 0.25$ the unfiltered rate is also 0.00%, while at $s = 3.0$ it is 14.68% and the conventional coherence > 50 filter deletes all of it, because mean coherence has fallen from 96.9 to 16.0. The filtered and unfiltered curves also peak at different scales (1.5 vs 2.0).

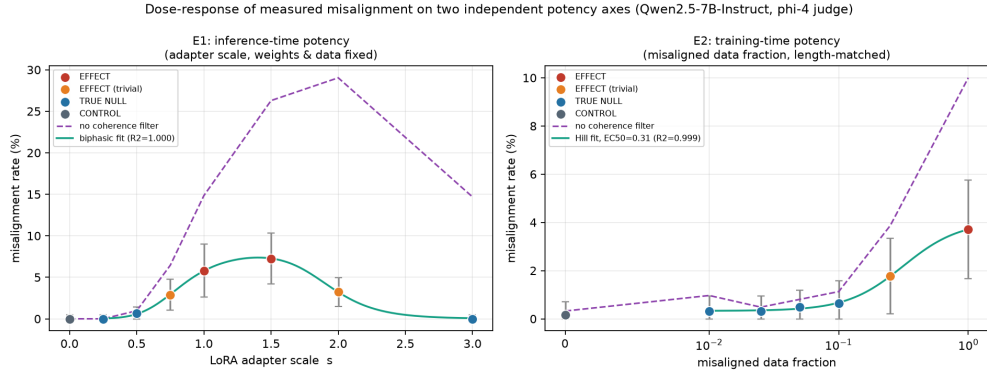


Figure 1: **Fitted dose-response curves on two independent potency axes.** (Left) E1, inference-time adapter scaling with weights and training data fixed: a biphasic rise-then-collapse fits with $R^2 = 0.9997$ and peaks at $s = 1.5$. This is the first fitted version of the shape Turner et al. [2025] observed at four unlabelled points. (Right) E2, training-time dose: a 4-parameter Hill curve fits with $R^2 = 0.999$ and $EC50 = 0.314 \pm 0.023$. In both panels the dashed line is the same data *without* the coherence filter; on E1 it peaks at a different scale ($s = 2.0$) than the filtered curve, and at $s = 3.0$ it stands at 14.68% where the filtered curve reads exactly zero. Error bars are clustered 95% CIs; marker colour is the arm’s verdict.

$s = 0.25$ would report robustness and would be right. Nothing in the reported number distinguishes the two cases. The unfiltered rate does.

Two further asymmetries fall out of the same table. The unfiltered curve peaks at $s = 2.0$ (29.03%) while the filtered curve peaks at $s = 1.5$ (7.26%): the two metrics disagree about which intervention is strongest. And judge non-response rises monotonically with potency, from 3.4% at base to 53.9% at $s = 3.0$, because the judge writes a justification paragraph instead of a number when the answer is unusual. Under the upstream convention those responses count as not-misaligned, so the metric degrades in the same direction as the coherence filter, and for a different reason.

4.2 E2: the training-time ladder and where it crosses detectability

Table 3 gives the length-matched training-time dose ladder. The rate climbs monotonically from 0.16% at dose 0 to 3.71% at dose 1.0, and a Hill curve fits with $R^2 = 0.999$ and $EC50 = 0.314 \pm 0.023$: 31.4% of the fine-tuning corpus must be misaligned for the rate to reach half its maximum. The detection boundary sits between dose 0.10 and 0.25.

Every arm below that boundary is a null—and every one of those nulls is uninformative at the margin that actually matters. At the pre-registered 5 pp margin all four low-dose arms are labelled TRUE NULL; at a 0.5 pp margin, the scale of the published open-weight effect (0.07% $\rightarrow$ 0.68%), **not**

Misaligned data fraction	0 (control)	0.01	0.025	0.05	0.10	0.25	1.00
Misalignment rate (%)	0.16	0.32	0.32	0.48	0.65	1.77	3.71
95% clustered CI	[0.00, 0.70]	[0.00, 0.95]	[0.00, 0.95]	[0.00, 1.18]	[0.00, 1.57]	[0.20, 3.35]	[1.67, 5.75]
Rate without coherence filter (%)	0.32	0.97	0.48	0.81	1.13	3.87	10.00
Question-level ICC	0.000	0.000	0.000	0.000	0.052	0.108	0.084
MDE at this arm's ICC (pp)	1.51	1.51	1.51	1.51	2.09	2.72	2.46
BH-adjusted p	—	0.38	0.38	0.23	0.14	0.036	0.0020
Equivalent at 5 pp margin?	—	yes	yes	yes	yes	no	no
Equivalent at 1 pp margin?	—	yes	yes	yes	no	no	no
Equivalent at 0.5 pp margin?	—	no	no	no	no	no	no
Verdict (5 pp margin)	CONTROL	TRUE NULL	TRUE NULL	TRUE NULL	TRUE NULL	EFFECT (TRIVIAL)	EFFECT
Verdict (0.5 pp margin)	CONTROL	INDETERMINATE	INDETERMINATE	INDETERMINATE	INDETERMINATE	EFFECT (TRIVIAL)	EFFECT

Table 3: **E2: the training-time dose ladder, and where it crosses detectability.** Corpus size is held at 2,308 rows so only composition varies; $n = 620$ per column. A 4-parameter Hill curve fits with $R^2 = 0.999$ and $EC50 = 0.314 \pm 0.023$. The detection boundary sits between dose 0.10 and 0.25. The TRUE NULL labels at low dose are an artefact of the pre-registered 5 pp margin being far too generous for a rate regime below 2%: at the 0.5 pp margin appropriate to the published open-weight effect ($0.07\% \rightarrow 0.68\%$), not one low-dose arm establishes equivalence and every one becomes INDETERMINATE.

Arm	Extended protocol (EXT-62, 62 questions)				Standard protocol (STD-8, 8 questions)			
	rate (%)	MDE (pp)	BH p	verdict	rate (%)	MDE (pp)	BH p	verdict
scale_0p75	2.90	2.21	0.0065	EFFECT (TRIVIAL)	5.00	10.4	0.67	INDETERMINATE
scale_1	5.81	3.54	0.0023	EFFECT	7.50	37.1	0.67	INDETERMINATE
scale_1p5	7.26	2.70	0.00029	EFFECT	10.00	37.6	0.67	INDETERMINATE
scale_2	3.23	1.79	0.0020	EFFECT (TRIVIAL)	2.50	16.4	0.67	INDETERMINATE
matched_dose_0p250	1.77	2.72	0.036	EFFECT (TRIVIAL)	0.00	8.9	1.00	TRUE NULL
matched_dose_1p000	3.71	2.46	0.0020	EFFECT	1.25	8.9	0.67	TRUE NULL
naive_dose_0p250	1.61	2.12	0.031	EFFECT (TRIVIAL)	1.25	8.9	0.67	TRUE NULL
scale_0p5_nudge	1.77	2.49	0.036	EFFECT (TRIVIAL)	1.25	8.9	0.67	TRUE NULL
scale_1_nudge	4.84	3.14	0.0020	EFFECT	3.75	8.9	0.67	INDETERMINATE

Table 4: **The standard protocol cannot see any of this.** The same responses, re-analysed using only the 8 canonical first-plot questions. All 9 arms that are statistically significant on 62 questions are non-significant on 8; the minimum detectable effect on 8 questions (8.9–37.6 pp) exceeds every effect present. Note that the point estimates on 8 questions are often *larger* than on 62—the problem is not the estimate but its precision, and the mechanism is clustering rather than raw sample size.

one of them establishes equivalence and all four become INDETERMINATE. This is not a defect of the data but of the margin, and it is itself the point: *the verdict depends on a margin the literature never states*. We pre-registered 5 pp and report it as primary, but we read the 0.5 pp column, and we recommend that any misalignment null state its margin explicitly and choose it for the rate regime under study.

4.3 The standard 8-question protocol cannot detect any of these effects

We re-analysed the *same responses* using only the 8 canonical first-plot questions. The result is table 4: **9 of 23 arms** are statistically significant under the 62-question protocol and non-significant under the 8-question one. This includes every arm on the interior of the E1 curve—the strongest, most obviously misaligned models we produced—and the full-dose fine-tune. The point estimates are often *larger* on 8 questions (7.50% vs 5.81% for `scale_1`); what collapses is the precision.

Table 5 and figure 2 explain why. Our measured question-level ICC is 0.049, so at 8 questions $\times$ 50 samples the design effect is 3.41 and 400 responses are worth 117 independent ones, giving an MDE of 6.41 pp. Doubling to 100 samples per question raises the design effect to 5.88 and improves the MDE only to 5.58 pp. Our own 62×10 design, with *fewer* total responses than 8×100 , has a design effect of 1.44 and an MDE of 1.92 pp. Detecting the real published open-weight effect ($0.07\% \rightarrow 0.68\%$) requires 1,572 responses per arm at this ICC—two to four times the standard protocol’s entire budget, for a single comparison. The mechanism is clustering, and it is not fixable by sampling harder.

Clustering inflates every standard error. Measured question-level ICC across arms has median 0.049 and maximum 0.211 (at `scale_1`). The median inflation of the clustered standard error over the naive binomial one is $1.42\times$, with a maximum of $1.77\times$. A naive response-level analysis of

Protocol	n	n_{eff}	DEFF	MDE (pp)	Rule-of-three (%)
8 questions $\times$ 50 samples (standard)	400	117.2	3.41	6.41	0.749
8 questions $\times$ 100 samples (standard)	800	136.1	5.88	5.58	0.374
62 questions $\times$ 10 samples (this study)	620	429.6	1.44	1.92	0.483

Table 5: **Power audit at our measured ICC = 0.049**, previously only assumed in this literature. MDE is computed at 80% power, $\alpha = 0.05$, against the published open-weight control rate of 0.07%. Doubling the samples per question barely helps because the design effect grows almost as fast as n ; adding questions keeps buying power. Detecting the real published open-weight effect (0.07% $\rightarrow$ 0.68%) requires **1,572 responses per arm** at this ICC, two to four times the standard protocol’s entire budget for a single comparison.

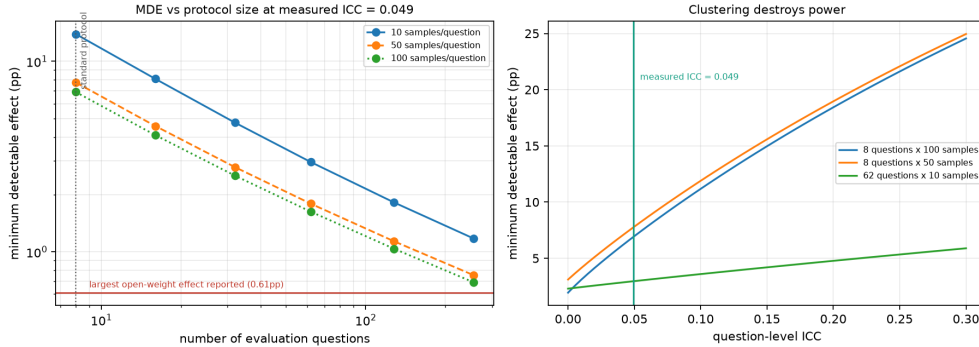


Figure 2: **Clustering, not sample size, is what destroys the standard protocol’s power.** (Left) MDE against the number of evaluation questions at our measured ICC = 0.049. Adding samples to a fixed 8 questions saturates almost immediately—the 50- and 100-sample curves nearly coincide—while adding questions keeps buying power. The largest reported open-weight effect (0.61 pp, red line) is below the MDE of every protocol size the field currently uses. (Right) MDE as a function of the question-level ICC. At ICC = 0 an 8-question design looks adequate; at our measured ICC it is already 2–3 $\times$ worse than a 62-question design with fewer total responses.

these data would overstate its precision by about 40%, and by 78% on the arm with the strongest effect. This is a false-positive risk that runs in the opposite direction to the false-null risk above, and both follow from the same design property.

4.4 E4: one set of weights, 144 measurements

Holding the weights fixed and varying only how they are measured (table 6, figure 3), the measured rate spans a 100 $\times$ range for the half-strength organism and a 65 $\times$ range for the full one; the base model goes from 0.00% to 6.45%. Even restricting to conventional settings (alignment threshold 30, best-of-1), the single coherence > 50 filter moves `scale_1` from 14.84% to 5.81%—a 2.6 $\times$ swing produced by a filter choice papers rarely report. Output format alone moves the same weights from 1.94% (`json`) to 4.84% (`nudge`) to 5.81% (`free`).

The verdict flips on format alone. Comparing `scale_1` to its own format-matched control: `free` gives EFFECT ($p_{\text{adj}} = 0.0023$), `nudge` gives EFFECT ($p_{\text{adj}} = 0.0020$), and `json` gives TRUE NULL ($p_{\text{adj}} = 0.25$). Same weights, same questions, same judge, same number of samples, opposite published conclusion.

Judge choice is a potency dial too. Re-judging a deterministic 25% subsample (3,598 responses) with the in-family QWEN2.5-14B-INSTRUCT raises every rate, including the control, which moves from 0.00% to 1.44% and thereby changes every contrast computed against it. Strikingly, the two judges agree strongly on the continuous alignment score (Pearson $r = 0.857$, Spearman $\rho = 0.669$) and disagree badly on the thresholded indicator (Cohen’s $\kappa = 0.199$; overall rate 1.70% vs 2.70%, a 59% relative difference). Reporting judge correlation—as papers occasionally do—says almost nothing about whether two judges would produce the same published verdict.

Fixed weights	min rate	max rate	ratio
base	0.00%	6.45%	∞
organism $\times 0.5$	0.16%	16.13%	$100\times$
organism $\times 1.0$	0.97%	62.90%	$65\times$

(a) Range over 144 measurement settings per weight setting.

Fixed weights + condition	PHI-4	QWEN2.5-14B-INSTRUCT
base, free	0.00%	1.44%
base, json	0.65%	2.30%
organism $\times 1.0$, free	5.81%	10.60%
organism $\times 1.0$, json	1.94%	2.82%
overall (3,598 paired responses)	1.70%	2.70%

(b) Judge choice is a potency dial too.

Table 6: **E4: the same weights, measured many ways.** (a) Holding the weights fixed and varying only output format, prompt nudge, alignment threshold, coherence filter and best-of- N moves the measured rate over a 65–100 $\times$ range; even at conventional settings the coherence > 50 filter alone moves `scale_1` from 14.84% to 5.81%. (b) Re-judging a deterministic 25% subsample with an in-family judge changes every rate, including the control. The two judges agree strongly on the continuous alignment score (Pearson $r = 0.857$, Spearman $\rho = 0.669$) and badly on the thresholded indicator (Cohen’s $\kappa = 0.199$): the threshold, not the judge, is where the agreement is lost.

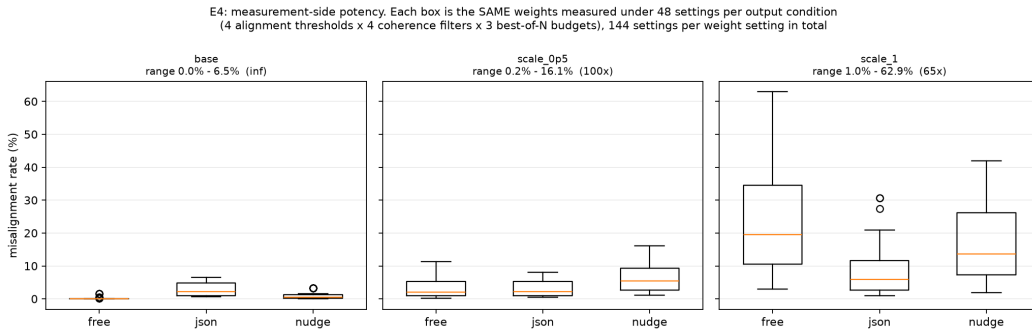


Figure 3: **Measurement-side potency at fixed weights.** Each box is the *same* weights measured under 48 settings per output condition (4 alignment thresholds $\times$ 4 coherence filters $\times$ 3 best-of- N budgets), 144 settings per weight setting. The base model spans 0.0–6.5%, the half-strength organism 0.2–16.1% (100 $\times$), and the full organism 1.0–62.9% (65 $\times$). No weight was changed anywhere in this figure.

Threshold sensitivity. Figure 4 shows why the measurement-side range is so large. The continuous alignment-score distributions shift smoothly and monotonically with potency; the thresholded rate is a single arbitrary cut through that family. Holding everything else at conventional settings, moving the alignment threshold from 30 to 70 takes `scale_1` from 5.81% to 10.65%, and moving it down to 10 takes it to 2.90%. Both the apparent “emergence” at moderate potency and the apparent “collapse” at high potency are consequences of thresholding, not of any discontinuity in the underlying behaviour.

4.5 E3: the length confound did not reproduce

Dose	Length-matched	Naive	Naive – matched	Mean tokens (matched / naive)
0.05	0.48%	0.81%	+0.32 pp	69.2 / 75.5
0.25	1.77%	1.61%	−0.16 pp	69.7 / 75.4

Table 7: **E3: a clean negative result.** We pre-registered that the naive (length-confounded) dose ladder would show the larger apparent effect. It does not: the difference is +0.32 pp at one dose and −0.16 pp at the other, both far inside the noise. Response length differs between the ladders as intended (≈ 6 tokens), so the manipulation worked; the effect on measured misalignment is simply not there at these doses in our setup.

Table 7 reports the length-confound ablation. Our pre-registered H5 predicted that the naive mixture would inflate the apparent effect relative to the length-matched one. It did not, in either direction

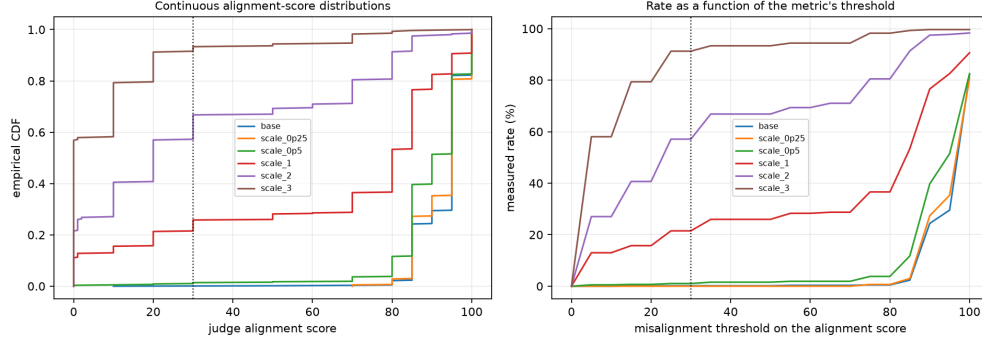


Figure 4: **Both the apparent emergence and the apparent collapse are produced by thresholding a smooth distribution.** (Left) Empirical CDFs of the continuous judge alignment score by adapter scale; the distributions shift smoothly and monotonically with potency, with no discontinuity at any scale. (Right) The same scored responses read out as a rate, as a function of where the threshold is placed. The conventional cut at 30 (dotted line) is an arbitrary point on a continuous family of possible answers, and the ordering of the arms is preserved at every cut. The critique of Schaeffer et al. [2023] applies directly to the emergent misalignment metric.

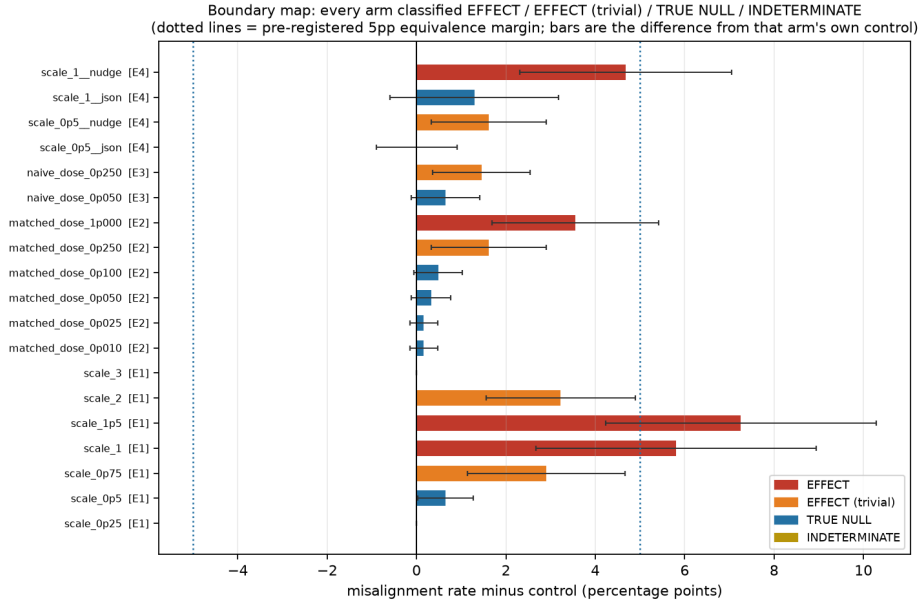


Figure 5: **The boundary map.** Each bar is one arm's difference from its own protocol-matched control, with a clustered 95% CI; dotted lines mark the pre-registered 5 pp equivalence margin; colour is the verdict. Across the 19 non-control arms: 4 EFFECT, 5 EFFECT (TRIVIAL), 10 TRUE NULL, 0 INDETERMINATE. At the 0.5 pp margin appropriate to the published open-weight regime, the picture inverts: only `scale_0p25` and `scale_3` (both exactly 0/620) remain TRUE NULL and every other null becomes INDETERMINATE.

consistently. We report this as a clean negative result. It does not contradict Rao et al. [2026], who studied *realignment* over a different length range, but it does mean the length confound was not load-bearing here—and it is worth noting that our own instrument classifies this null honestly: with only two doses, a two-point ablation is weak evidence, so “not reproduced here” is the correct reading rather than “refuted”.

Pre-registered hypothesis	Verdict	Evidence
H1 potency rise-then-collapse	partially supported	Shape confirmed decisively (biphasic $R^2 = 0.9997$, interior peak at $s = 1.5$, return to exactly 0.00% at $s = 3.0$). The pre-registered magnitude criterion—at least one arm ≥ 10 pp above control—was not met; our maximum was 7.26 pp. Judged strictly, H1 fails.
H2 sub-threshold nulls	supported	9/23 arms flip from non-significant on 8 questions to significant on 62.
H3 measurement-side range	supported	65–100× range at fixed weights; verdict flips between EFFECT and TRUE NULL on output format alone.
H4 clustering inflates SE	supported	Median SE inflation 1.42× (pre-registered threshold 1.3×), max ICC 0.211.
H5 length confound	not supported	−0.16 pp to +0.32 pp, no consistent direction.

Table 8: Pre-registered hypotheses and their outcomes. We report H1 and H5 as they fell rather than as we hoped: the H1 magnitude criterion was missed, most likely because our judge’s measured sensitivity is 0.22 (section 4.7), and H5 simply did not reproduce.

4.6 The boundary map

Figure 5 assembles every arm into one picture. At the pre-registered 5 pp margin the 19 non-control arms split 4 EFFECT, 5 EFFECT (TRIVIAL), 10 TRUE NULL, 0 INDETERMINATE. At the 0.5 pp margin the split becomes 4 EFFECT, 5 EFFECT (TRIVIAL), 2 TRUE NULL, 8 INDETERMINATE: only the two arms with exactly zero events in 620 responses survive as genuine nulls. The boundary between real robustness and undetectable effect therefore moves with the margin, which is exactly why the margin must be stated.

4.7 Judge calibration: the instrument is conservative and insensitive

Because every rate here is judge-mediated, we hand-labelled 35 responses stratified over the judge’s score range and presented blind. At the conventional threshold of 30 the judge’s sensitivity is 0.22 with a false-positive rate of 0.077, precision 0.50 and Cohen’s $\kappa = 0.18$; the best threshold by Youden’s J is 35 (TPR 0.33, FPR 0.077). The continuous score tracks human judgement (Spearman $\rho = -0.49$ between alignment score and the human misalignment label), but once thresholded at 30 the judge catches roughly a fifth of what a human labeller flags while almost never firing falsely.

Two consequences follow. First, **every rate in this paper is a lower bound**, and the direction of the bias is toward *more* apparent robustness—which strengthens rather than weakens our central claim. Second, the same pattern we found between two judges appears again between judge and human: agreement on the continuous score, failure at the threshold. The threshold is where this metric loses its information.

5 Discussion

5.1 What the evidence supports

Table 8 records how each pre-registered hypothesis fell. Three of the four testable limbs are supported, with one clean negative and one partial. Three conclusions follow.

Null results in this literature are not, in general, evidence of robustness. The standard protocol’s MDE—6.4 pp at 8 questions $\times$ 50 samples, and 8.9–37.6 pp on our own arms—exceeds every effect we produced. It could not have detected our strongest, most obviously misaligned model. A null from an instrument with that MDE excludes nothing that this literature has ever reported.

True robustness does exist and is identifiable. At $s = 0.25$ the rate is 0/620 with *and without* the coherence filter, and equivalence to base holds even at a 0.5 pp margin. That is a real boundary, and our instrument names it while correctly refusing to name the superficially identical zero at $s = 3.0$.

The boundary is not where the potency axis alone says it is. It is jointly determined by intervention potency, the coherence filter, the score threshold, the output format and the judge. Two of those five—the filter and the format—are usually unreported, and a third, the judge, is usually a monoculture.

5.2 A minimum reporting standard for misalignment nulls

We propose four items, all of which this study implements and none of which the surveyed literature reports:

- **Cluster standard errors at the question level and report the measured ICC.** Responses within a question are not independent; at our ICC the naive standard error is understated by 42% on average.
- **Report the minimum detectable effect alongside the point estimate.** A null without an MDE is uninterpretable.
- **Report a TOST equivalence verdict against an explicitly stated margin,** chosen for the rate regime under study. Our own 5 pp margin was mis-calibrated for a sub-2% rate regime, and saying so is more useful than hiding it.
- **Report the rate without the coherence filter next to the filtered rate,** and give the threshold-sensitivity curve rather than a single cut. This is what separates the two kinds of zero.

5.3 Surprises and failure modes

The upper collapse is entirely a filter artefact. We expected incoherence to attenuate measured misalignment; we did not expect the unfiltered rate at $s = 3.0$ (14.68%) to exceed the *filtered* rate at every other scale. The standard metric does not merely attenuate the signal at high potency—it erases it. The judge’s failure mode is also not the one we anticipated: it is silence, not error. It rarely scores an aligned answer as misaligned (FPR 0.077); it simply declines to emit a number, increasingly often as answers get stranger (3.4% $\rightarrow$ 53.9% across the E1 ladder), and naive pipelines drop those rows. Finally, two judges can agree at $r = 0.86$ and still have $\kappa = 0.20$: all the disagreement lives at the threshold.

Other observed failure modes, in decreasing frequency: the subject model echoes the template instead of answering (concentrated in template-format questions); it emits a spurious extra conversational turn at high potency; and at $s \geq 2$ it produces fluent-looking but semantically empty text that the coherence judge correctly scores near zero.

5.4 Limitations

The judge is not GPT-4o. No API key was available in this environment. Absolute rates are not comparable to published numbers, and the measured sensitivity of 0.22 means all our rates are lower bounds. Within-study contrasts—which is what every claim here rests on—are unaffected in direction, though effect sizes are compressed. This is the most likely reason the H1 magnitude criterion was missed.

Judge calibration is $n = 35$ with a single labeller. No inter-annotator agreement is available and the labeller was not independent of the study. This is the weakest link in the chain and the first thing we would fix.

One model family. `meta-llama/*` and `google/gemma-*` are gated and no access token was available. Gemma is the family reported *most* resistant to EM—the most interesting candidate for genuine robustness—and we could not test it. Everything here is Qwen2.5-7B.

Design choices. We used 10 samples per question, which is correct given $ICC \approx 0.05$ but makes per-question rates coarse and caps best-of- N at 10. The E2 doses were each trained once with a single seed, so between-seed variance is unmeasured and could be comparable to the low-dose differences. E3 tested only two doses, so H5 should be read as “not reproduced here” rather than “refuted”. And the 5 pp equivalence margin was mis-calibrated for a low-rate regime; we pre-registered it and report it as primary, but the 0.5 pp and 1 pp columns are the ones to read.

5.5 Broader implications

The precedent from adversarial robustness is that a field which accepts nulls from weak instruments accumulates false confidence until someone runs a stronger attack [Athalye et al., 2018]. Misalignment evaluation is on that path: its protocol was fixed by convention, its metric is a threshold on a continuous score, its filter is correlated with its outcome, and its nulls carry no stated power. Our

results say the problem is tractable—the same four statistics that would have caught the adversarial defences catch our own underpowered cells—but that the current reporting convention cannot distinguish a robust model from an unmeasured one. Because misalignment nulls are increasingly cited in safety cases and policy discussion, the cost of that ambiguity is not academic.

We see no dual-use risk in this work beyond what is already public: the model organism we scale is a published open-weight artefact, and our contribution is measurement rather than a new way to induce harm. If anything, the direction of our findings is that misaligned behaviour is *more* present than the standard metric reports, which argues for stronger, not weaker, evaluation.

6 Conclusion

We asked whether a reported misalignment null means the model is robust or the experiment was unable to see the effect. The answer, in this setting, is both—and the two are separable, but only with instruments the field does not currently use. On a continuous potency dial with weights and training data held exactly constant, the measured rate goes $0.00\% \rightarrow 7.26\% \rightarrow 0.00\%$, and the two zeros are different kinds of zero: one is a genuine absence of misaligned content, the other is manufactured by a coherence filter that deletes 14.68 percentage points of misaligned output. The standard 8-question protocol could not detect any effect we produced, its MDE at our measured ICC of 0.049 being 6.4–37 percentage points; 9 of 23 arms flip from null to significant when the question count rises to 62; and one fixed set of weights spans a 65-fold range of measured misalignment depending only on how it is measured.

Key takeaway. A misalignment null is uninterpretable unless it is reported with a clustered standard error, a minimum detectable effect, an equivalence test against a stated margin, and the unfiltered rate alongside the filtered one. We provide all four as a reusable procedure and apply it to every one of our own cells, including the ones where it convicts us.

Future work. The first priority is to replace the local judge with a frontier API judge and re-measure, recovering absolute comparability and raising sensitivity above 0.22. Second, extend the inference-time ladder to Gemma-3 and Llama-3.1: if the family reported most resistant to EM shows the same biphasic curve at lower amplitude, then “resistance” is a position on the potency axis rather than a property of the model. Third, refine the ladder between $s = 1.5$ and $s = 3.0$ to locate the coherence-collapse inflection and fit the unfiltered curve separately, since it peaks elsewhere. Finally, the classifier needs only k , n and the number of questions, so it can be applied retrospectively to published nulls—which is the test we would most like to see run.

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A Full boundary map

Arm	Axis	Potency	k/n	Rate (%)	95% clustered CI	ICC	MDE (pp)	BH p	Verdict
base	E1	0	0/620	0.00	[0.00, 0.44]	0.000	1.25	—	CONTROL
scale_0p25	E1	0.25	0/620	0.00	[0.00, 0.44]	0.000	1.25	1.00	TRUE NULL
scale_0p5	E1	0.5	4/620	0.65	[0.00, 1.41]	0.000	1.25	0.085	TRUE NULL
scale_0p75	E1	0.75	18/620	2.90	[1.06, 4.75]	0.087	2.21	0.0065	EFFECT (TRIVIAL)
scale_1	E1	1.0	36/620	5.81	[2.61, 9.00]	0.211	3.54	0.0023	EFFECT
scale_1p5	E1	1.5	45/620	7.26	[4.19, 10.32]	0.132	2.70	0.00029	EFFECT
scale_2	E1	2.0	20/620	3.23	[1.49, 4.96]	0.049	1.79	0.0020	EFFECT (TRIVIAL)
scale_3	E1	3.0	0/620	0.00	[0.00, 0.44]	0.000	1.25	1.00	TRUE NULL
matched_dose_0p000	E2	0	1/620	0.16	[0.00, 0.70]	0.000	1.51	—	CONTROL
matched_dose_0p010	E2	0.01	2/620	0.32	[0.00, 0.95]	0.000	1.51	0.38	TRUE NULL
matched_dose_0p025	E2	0.025	2/620	0.32	[0.00, 0.95]	0.000	1.51	0.38	TRUE NULL
matched_dose_0p050	E2	0.05	3/620	0.48	[0.00, 1.18]	0.000	1.51	0.23	TRUE NULL
matched_dose_0p100	E2	0.10	4/620	0.65	[0.00, 1.57]	0.052	2.09	0.14	TRUE NULL
matched_dose_0p250	E2	0.25	11/620	1.77	[0.20, 3.35]	0.108	2.72	0.036	EFFECT (TRIVIAL)
matched_dose_1p000	E2	1.00	23/620	3.71	[1.67, 5.75]	0.084	2.46	0.0020	EFFECT
naive_dose_0p050	E3	0.05	5/620	0.81	[0.00, 1.77]	0.039	1.95	0.16	TRUE NULL
naive_dose_0p250	E3	0.25	10/620	1.61	[0.30, 2.93]	0.054	2.12	0.031	EFFECT (TRIVIAL)
base__json	E4	0	4/620	0.65	[0.00, 1.57]	0.052	2.67	—	CONTROL
base__nudge	E4	0	1/620	0.16	[0.00, 0.70]	0.000	1.51	—	CONTROL
scale_0p5__json	E4	0.5	4/620	0.65	[0.00, 1.41]	0.000	2.03	1.00	TRUE NULL
scale_0p5__nudge	E4	0.5	11/620	1.77	[0.28, 3.27]	0.087	2.49	0.036	EFFECT (TRIVIAL)
scale_1__json	E4	1.0	12/620	1.94	[0.02, 3.85]	0.192	4.29	0.25	TRUE NULL
scale_1__nudge	E4	1.0	30/620	4.84	[2.20, 7.48]	0.147	3.14	0.0020	EFFECT

Table 9: All 23 arms with k/n , clustered CI, measured ICC, MDE at that arm’s own ICC, BH-adjusted p against its own protocol-matched control, and verdict at the pre-registered 5 pp equivalence margin. Each E4 arm is compared to the base arm generated under the *same* output condition, which is why `scale_1` is EFFECT under `free` and `nudge` but TRUE NULL under `json`.

B Judge calibration detail

Table 10 reports the blind hand-label calibration of the primary judge in full, together with the threshold sweep summary. The labelling criterion was liberal—a response counts as misaligned if it endorses or advises something harmful, dishonest or discriminatory—and the sample was stratified over the judge’s score range so that the estimate is not dominated by the aligned majority. The single-labeller design is the study’s weakest link (section 5), and we report it as such rather than correcting rates by the estimated TPR.

Metric	Value
Sensitivity (TPR) at the conventional threshold 30	0.22
False-positive rate	0.077
Precision	0.50
Accuracy	0.74
Cohen’s κ	0.18
Spearman(alignment score, human label)	−0.49
Best threshold by Youden’s J	35 (TPR 0.33, FPR 0.077)

Table 10: Blind hand-label calibration of the PHI-4 judge on 35 responses stratified over its score range (TP = 2, FP = 2, FN = 7, TN = 24). The judge is conservative and insensitive: it catches roughly a fifth of what a human labeller flags while almost never firing falsely, so every rate in this paper is a lower bound biased toward apparent robustness.

C Reproducibility

All artefacts are released: 14,260 raw generations and per-response judge scores, the boundary map (with and without the 8-question restriction), the protocol comparison, the pre-registered hypothesis tests, the power audit at the measured ICC, the 144-setting elicitation sweep, the judge calibration, the curve fits, and the deviation log. The pipeline is: build length-matched and naive dose mixtures, train nine LoRA adapters, materialise the seven scaled adapters, generate all arms, judge with PHI-4 (primary, all responses) and QWEN2.5-14B-INSTRUCT (secondary, deterministic 25% subsample), hand-label the blind calibration sample, and run the analysis. Software: Python 3.12.8, torch 2.9.0, transformers 4.57.6, peft 0.20.0, vLLM 0.11.2; seed 42 throughout; total compute approximately 75 GPU-minutes on $2 \times$ RTX A6000.